

Complete Raw-Text Neural Speech Synthesis on an ESP32-P4

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Abstract

This paper describes the deployment of a complete English text-to-waveform system on an ESP32-P4 microcontroller. The device accepts raw UTF-8 text and executes normalization, eSpeak-NG phonemization, model-specific tokenization, an acoustic model, duration and monotonic-path construction, latent sampling, inverse normalizing flow, neural waveform decoding, and PCM16 conversion. No host processor contributes phonemes, tokens, latent tensors, or waveform samples. The evaluated board contains an early-revision dual-core 360 MHz ESP32-P4, 32 MiB of in-package PSRAM, and 16 MiB of flash; it has no neural processing unit.

The port uses the 3.96-million-parameter Inflect-Nano-v2 model and ESP-DL. The main deployment work comprised exact polyphase replacement of unsupported transposed convolutions, time-axis dual-core scheduling for Conv1D, layout specialization and graph passes, mixed INT8/INT16 quantization, resident model objects, static flow and decoder length buckets, and an on-device text segmentation planner. The first complete waveform decoder required 12.809 s for 1.525 s of audio. The final short-utterance route decodes a 44-frame request in 1.335 s and executes all learned stages in 1.656 s, producing 0.469 s of 24 kHz speech. The board retains 17.54 MB of free PSRAM and approximately 99.6 kB of free internal RAM after the frontend and decoder routes have been exercised. Byte counts, model digests, board and host checksums, tensor parity tests, and direct listening were used as separate acceptance gates. The current system is useful for short device speech and bounded queued announcements. Its exact decoder remains slower than real time and does not expose native streaming state.

1 Introduction

Neural text-to-speech systems are usually described by model size and host runtime. Those measurements omit several costs that become decisive on a microcontroller: text normalization, grapheme-to-phoneme conversion, static tensor shapes, unsupported operators, quantization metadata, flash layout, temporary stacks, audio transport, and the lifetime of multiple model objects. A waveform decoder that runs on a board proves only one part of a speech system. A device that receives ordinary text has to execute the entire path and return a verifiable waveform.

This work ports Inflect-Nano-v2 [13], a fixed-voice English model derived from the VITS family [9], to an ESP32-P4. The release contains fewer than four million parameters and includes its waveform generator. Its size makes a complete deployment plausible, although its HiFi-GAN-like decoder [10] still performs billions of multiply-accumulate operations per second of output audio. The target board has generous PSRAM for a microcontroller, yet memory bandwidth and internal SRAM remain limited. It also lacks a GPU or NPU. The available acceleration comes from dual RISC-V cores, ESP32-P4 processor extensions, cache, and the quantized kernels in ESP-DL [3, 4].

The completed system accepts a raw text command. The board normalizes the text, invokes a pinned eSpeak-NG frontend, maps phonemes into the released symbol order, inserts VITS blanks, executes every learned inference stage, and converts the final tensor to 24 kHz mono PCM16. USB supplies power and a temporary text/debug channel during the experiment. Replacing that channel with a local application input and an I2S sink requires no inference on an external computer.

Figure 1 shows the experimental USB connection and the intended standalone boundary. The development host supplies text and power and may capture diagnostic PCM, while every speech computation remains on the microcontroller.

The work makes four contributions.

- It documents a complete raw-text neural TTS deployment on an ESP32-P4, including the frontend, stochastic latent construction, flow, decoder, and audio conversion.

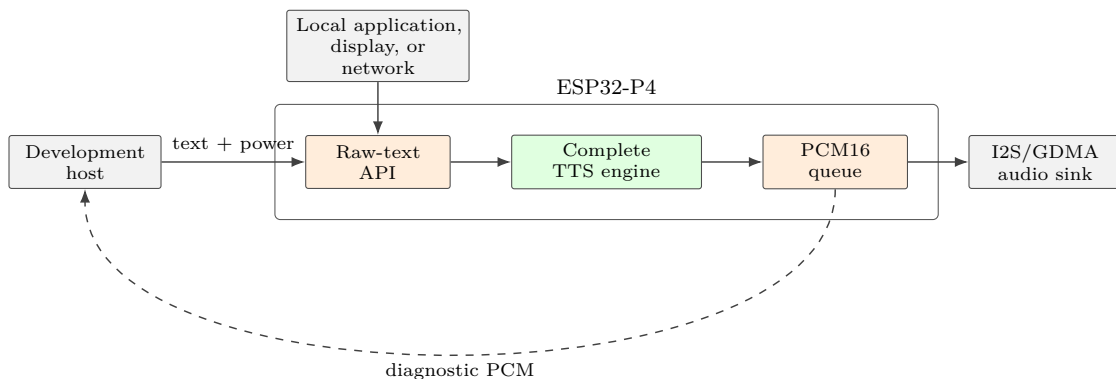


Figure 1: System boundary. USB is a development transport and power source; phonemes, tokens, latent tensors, and waveform samples are produced inside the ESP32-P4. A standalone application can call the same raw-text API and send PCM to an I2S/GDMA sink.

- It gives measured implementations for model and runtime problems that are hidden by parameter count: lowering transposed convolution, Conv1D scheduling, repeated layout conversion, mixed precision, and fixed-shape routing.
- It defines an evidence procedure that separates graph execution, integer parity, transport integrity, and audible speech quality. Several faster candidates had valid checksums and failed listening tests. The gates therefore remain separate.
- It reports flash, PSRAM, internal-memory, stack, latency, and static shape limits for a reproducible board and software revision.

The measured exact-weight runtime is slower than real time. This paper makes no conversational or native-streaming claim. The practical result is a complete local speech engine for short values, menu items, status messages, navigation cues, and bounded queued announcements. The same port also gives a measured starting point for later decoder training and causal synthesis work.

2 Related Work

2.1 End-to-end neural speech synthesis

VITS combines a text encoder, duration modeling, variational latent variables, normalizing flows, and adversarial waveform generation in a single training system [9]. Inflect follows this broad inference structure and packages a complete fixed-voice text-to-waveform model. Its release API supports CPU, CUDA, and ONNX execution, but the published deployment interface returns complete waveform chunks and has no streaming state [14, 15].

The waveform generator belongs to the family popularized by HiFi-GAN, where transposed convolutions increase temporal resolution and multi-receptive-field residual blocks refine the signal [10]. Such generators are parallel on server hardware. Their long low-channel convolutions can be a poor fit for an MCU because activations grow to waveform rate while cache and SIMD width remain constrained.

Alternative vocoders reduce this cost through model design. LPCNet combines linear prediction with a sparse recurrent network [18], and later work reduced its cost further [16]. FARGAN generates small subframes and uses long-term pitch prediction at a reported 600 MFLOP/s [17]. Vocos predicts Fourier coefficients instead of constructing a waveform through a deep time-domain upsampler [12]. MB-iSTFT-VITS replaces part of a VITS waveform path with inverse STFT and reports a 4.1-fold CPU synthesis speedup in its setting [8]. These designs provide directions for a future trained decoder. They cannot replace the released Inflect weights without a new training and quality evaluation cycle.

2.2 Neural inference on microcontrollers

MCUNet showed that architecture and runtime have to be designed together under MCU memory constraints [11]. TinyVocos applies this idea to vocoding: its authors evaluated several purpose-built models on four MCUs and

Table 1: Measured platform and pinned software.

Item	Exact state
SoC	ESP32-P4 revision 1.3, two high-performance RISC-V cores at 360 MHz
Memory	32 MiB in-package PSRAM, hexadecimal mode, 200 MHz
Flash	16 MiB, 80 MHz DIO in the tested configuration
Inference SDK	ESP-IDF 6.0.2 and ESP-DL 3.3.9
Exporter	ESP-PPQ 1.3.7, ONNX 1.17.0, Python 3.12.11
Model	<code>owensong/Inflect-Nano-v2</code> , revision <code>1cbf8079...</code>
Checkpoint	15,971,083 B; SHA-256 <code>bfca4684...25b3</code>
Frontend	eSpeak-NG 1.52.0, commit <code>4870adfa...d6c</code>
Output	24 kHz, mono, signed PCM16 little-endian

Table 2: Learned inference decomposition. Parameter counts refer to the active deployment graph after weight-norm removal.

Part	Parameters	Primary shape	Function
Embedding	12,816	178×72	Token ID to quantized text embedding
Acoustic core	837,297	$T_{text} \leq 49$	Text encoding, statistics, and durations
Inverse flow	1,016,320	$128 \times T_z$	Prior latent to decoder latent
Waveform decoder	2,092,368	$128 \times T_z$	256 PCM samples per latent frame

report a best footprint of 1.5 MiB flash and 517 KiB RAM with MOS 3.95 [1]. TinyVocos receives acoustic features and does not provide the complete raw-text path evaluated here, but it establishes that a modern neural vocoder can fit MCU-class resources.

ESP-DL provides quantized operators, a static memory planner, FlatBuffer-based model files, and dual-core execution for Espressif chips [3]. Its general operator set made the current port possible. The installed implementation was oriented toward Conv2D scheduling; the time dimension of a rank-three Conv1D tensor did not use both cores. The runtime modifications described in Section 4.4 address the specific shapes of this speech decoder.

3 Target System

3.1 Hardware and software freeze

All board measurements were taken on one connected ESP32-P4 revision 1.3. This early revision was configured at its supported 360 MHz frequency. The module exposes 32 MiB of in-package hexadecimal PSRAM at 200 MHz and 16 MiB of external flash. Table 1 gives the state used for the final route. ESP32-P4 includes processor extensions and image/audio peripherals, but it does not contain a neural processing unit [6].

The software state was frozen before graph changes. The Hugging Face snapshot, model checkpoint, configuration, ESP-DL component, exporter environment, and host golden tensors were identified by commit or SHA-256. Version pinning was necessary because graph layout and per-channel quantization behavior changed across ESP-DL and ESP-PPQ releases. Historical exports made with ESP-PPQ 1.3.6 remain reproducible evidence; new final-route exports use 1.3.7.

3.2 Model decomposition

The checkpoint contains 3,966,721 parameters before release-time weight-norm removal and 3,958,801 active parameters after removal. The waveform decoder contains 2,092,368 active parameters. We split inference into an embedding table, an acoustic graph, deterministic MCU glue, an inverse-flow graph, and a waveform decoder. Table 2 records the learned parts.

Figure 2 gives the complete data path and distinguishes learned ESP-DL graphs from deterministic firmware stages. This distinction was required for memory accounting and parity tests.

At runtime, normalized text becomes phoneme tokens x . The acoustic graph returns latent statistics and log du-

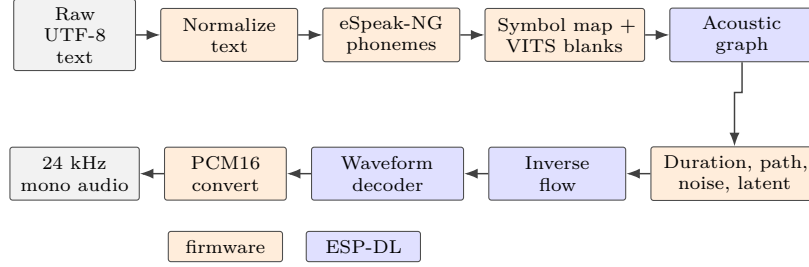


Figure 2: Complete on-device text-to-waveform path. Orange blocks are explicit firmware or the embedded eSpeak frontend; blue blocks are learned graphs executed by ESP-DL. USB does not perform any stage in this path.

rations. Firmware applies the released duration exponentiation, speed scaling, ceiling, and bounds, then constructs the monotonic token-to-latent path. Means and log scales are expanded along that path. A pinned standard-normal noise tensor, seed, and variation control produce the prior latent z_p . Four inverse coupling blocks map z_p to the decoder latent z . The waveform generator expands each latent frame by a factor of 256 and returns a mono waveform.

Several operations remain in C++ instead of a learned graph. Embedding lookup is a bounded table access. Mask construction, duration rounding, monotonic path construction, latent expansion, noise combination, tensor-layout conversion, PCM conversion, and checksums are simpler and smaller as explicit firmware code. A local floating-point **Where** module fills one gap in the installed ESP-DL operator set.

3.3 Waveform decoder structure

The active release decoder receives $[B, 128, T_z]$. A 7-tap convolution maps 128 channels to 192. Four transposed convolutions then upsample by 8, 8, 2, 2 while reducing channels to 96, 48, 24, 12. Each stage averages three residual branches with kernel sizes 3, 7, and 11; each branch uses dilation pairs derived from 1, 3, and 5. A final leaky activation, 7-tap convolution, and hyperbolic tangent produce the waveform. The graph contains 74 ordinary Conv1D operations, four ConvTranspose1D operations, and twelve residual blocks.

For a 96-frame input, the decoder performs 4,159,143,936 nominal MACs and emits 24,576 samples, or 1.024 s. This is 4.062 GMAC per second of output audio. The $k = 11$ residual branches account for 50.54% of decoder MACs; $k = 7$ branches account for 32.16%, and $k = 3$ branches account for 13.78%. The pre/post convolutions and four upsamplers together account for about 3.52%. The decoder is small in parameter bytes and expensive in repeated time-domain arithmetic.

An analytic dependency trace and direct zero-padding checks found 13 latent frames of context on each side of an interior output region. The complete dependency span is 27 frames. Since each frame represents 256 samples, one side of context equals

$$\frac{13 \cdot 256}{24000} = 138.667 \text{ ms.} \quad (1)$$

This context determines safe bucket routing and rules out zero-lookahead exact streaming with the released weights.

4 Porting Method

4.1 Reference-first procedure

The port was developed as a sequence of falsifiable stages. A deterministic host canary, **Testing one two three.** with seed 17, produced input tokens, the latent entering the decoder, intermediate tensors, the float waveform, and hashes. Operator feasibility was measured before the complete graph was exported. Each graph rewrite had to pass a float comparison before quantization, the ESP-PPQ host executor after quantization, and the ESP-DL embedded test vector on the board.

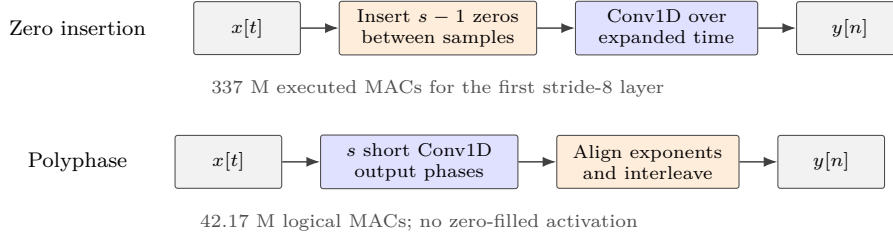


Figure 3: Exact polyphase lowering of transposed convolution. The original export expanded time with zeros before convolution. The deployed graph computes one short convolution per output phase and interleaves the results.

The first real-layer gate used all 27,456 outputs of the decoder pre-convolution. It exposed two deployment details. ESP-PPQ packs P4 filters in 16-channel blocks rather than ordinary contiguous PyTorch order, and the P4 kernel uses round-half-to-even during accumulator shifts. Implementing both rules produced zero integer mismatches for per-tensor and per-channel weight quantization. This layer gate prevented graph-level audio errors from being misdiagnosed as model-quality problems.

4.2 Exact polyphase transposed convolution

The initial ESP-PPQ lowering of `ConvTranspose1d` inserted zeros before an ordinary convolution. For the first $k = 16$, stride-8 upsampler, this executed about 337 million MACs in place of the 42.17 million logical MACs. The upsample convolution alone took about 2.6 s on the board.

We replaced every transposed convolution with an exact polyphase form. A stride- s transposed convolution can be separated into s ordinary convolutions, one for each output phase. The phase outputs are interleaved in time after any exponent alignment. The two $k = 16$, stride-8 layers use eight two-tap phases; the two $k = 4$, stride-2 layers use two two-tap phases. No zero-filled activation is materialized.

Figure 3 contrasts the exported zero-insertion graph with the exact implementation used on the board. Both routes implement the same linear operator; the second executes only the nonzero phases.

The complete transformed FP32 decoder differed from the release decoder by 5.49×10^{-8} mean absolute error and 2.97×10^{-6} maximum absolute error on the fixed latent. Correlation was 0.999999999996. On the first complete residual stage, polyphase lowering reduced board time from 3.493 s to 1.366 s, a 2.557-fold speedup, with zero integer mismatches after quantization.

The initial implementation built the interleave from portable slice, concat, reshape, and transpose operators. A later custom fused interleave was measured on all four real shapes. It reduced that layout chain from 101.120 ms to 78.967 ms. The projected whole-decoder gain was only 0.772%, so the custom operator was not included in the final graph. This test prevented a visually large graph rewrite from consuming further effort after its measured upper bound became small.

4.3 Quantization and audible output precision

ESP-PPQ exports power-of-two quantization compatible with ESP-DL. Convolution weights use symmetric per-output-channel scales, while ordinary activations use symmetric per-tensor scales. Integer-only inference follows the usual affine-quantization principles [7], with the target’s packing and rounding behavior included in parity tests. The exporter version and its target-specific dispatch are pinned separately from the firmware runtime [5].

The first complete decoder used INT8 throughout and returned an INT8 waveform with exponent -8 . Its WAV container was PCM16, although every sample was a multiple of 128. A valid board utterance used only 121 distinct sample levels. Direct listening found stable background noise. Quantizing only a clean FP32 waveform to the same final step gave 35.29 dB SNR and 0.999852 correlation; the complete INT8 graph gave 4.11 dB SNR and 0.789805 correlation. Error had already accumulated before the final output quantizer.

Layerwise analysis identified the input convolution and first upsampling stage as sensitive. Three mixed-precision candidates were exported. Promoting individual residual convolutions introduced a precision boundary inside a residual path and worsened error. Promoting the input convolution and first upsampling pair improved correlation from 0.790 to 0.824. Extending INT16 to the final convolution and tanh produced an INT16 output at exponent -16

with 6,120 distinct levels and a nonzero-value greatest common divisor of 1 on the board. This **mixed-front-head** route increased the 96-frame decoder time by 6.1% relative to all INT8. It was retained because direct listening preferred its usable output precision and because the remaining memory was ample.

Waveform statistics were treated as diagnostics. Correlation, SNR, RMS, clipping, and unique levels can locate a quantization failure; none establishes intelligibility or naturalness. The experiment therefore allowed a changed PCM hash after a declared quantization change and required direct listening to the exact board artifact.

4.4 ESP-DL runtime changes

Inspection of the installed ESP-DL source found that multi-core Conv2D divided work along input height. Conv1D tensors were represented with height one and time along width, so this path could not divide speech convolutions between cores. A narrow Conv1D path now splits the output time axis. Each core receives one output interval and only the input overlap required by the dilated kernel. Outer padding is applied at the corresponding end. Quantization metadata ownership remains with the original runtime argument.

At 512 KiB L2 cache, the custom single-core decoder took 8.630 s and the dual-core path took 5.037 s, a 1.713-fold gain with identical 36,608-byte INT8 output. The gain is below two because both cores share the PSRAM/cache path and some operators remain serial.

The stock graph also spent seconds moving between channel-first and channel-last layouts. A specialized transpose handles the common $[0, 2, 1]$ permutation as tiled contiguous copies and divides large copies between cores. Two graph passes then reduced the number of transfers. The first deduplicates post-layout transpose fan-outs only when shape, data type, exponent, layout, permutation, and target platform match. It removed twelve operations and reduced a padded decoder by 142.4 ms. The second propagates native NWC layout through compatible slice paths and removed eight more transpose nodes.

Late decoder stages use 24 and 12 channels, which underfill the P4 vector kernels. Padding these stages to 32 and 16 channels, while maintaining zeros in added channels across convolution, activation, residual addition, and output cropping, reduced the decoder from 5.578 s to 3.971 s. The output remained byte-identical. This trade spent about 1.25 MiB of additional PSRAM to reduce execution by 28.8%.

Activation profiling showed 69 leaky-ReLU lookup tables. Exhaustive inspection of every signed INT8 input reduced them to seven exact formula classes. A P4 SIMD implementation applies the corresponding scale, negative slope, round-half-to-even shift, and small correction masks for decimal slopes. All 138 tables across two decoder buckets matched for all 256 inputs. The formula path reduced the full decoder by 42.3 ms and the 96-frame decoder by 27.4 ms. Unrecognized tables keep the generic ESP-DL lookup implementation.

Several apparently useful changes were rejected by direct board A/B tests. Dual-core flat additions and requantization increased PSRAM traffic and slowed the graph. A larger ESP-DL tensor-planner allocation consumed scarce internal RAM for less than a 0.3% latency change. Executing parameters from flash saved 2.13 MiB PSRAM but made inference 27.4% slower. These results illustrate why operator count and isolated kernel speed did not predict the final runtime.

4.5 Resident service and model buckets

Repeated model construction added more than one second to each request. The command service verifies the model package once at boot, constructs model objects on first use, and keeps them resident. A historical 143-frame route dropped from 5.336 s for the first request to 3.789 s when warm, while the learned arithmetic remained near 3.53 s.

Static decoder shapes wasted work on zero padding. A 143-frame decoder took about 2.96 s whether a phrase required 71, 74, or 106 frames. We exported a 96-frame decoder and proved that its output equals the first 83 valid frames of a zero-padded 143-frame graph. The remaining 13 frames are right context. For **Good morning.**, selecting T96 reduced the decoder from 2.962 s to 2.073 s. The later accepted mixed-precision and runtime path measured about 2.013 s.

The final short-value package adds a T64 decoder. T64 has a 51-frame safe prefix after reserving right context. The measured **Ready.** request requires 44 frames, so T64 replaces T96 for this case and reduces decoder time from 2.013 s to 1.335 s. The complete residual structure remains intact.

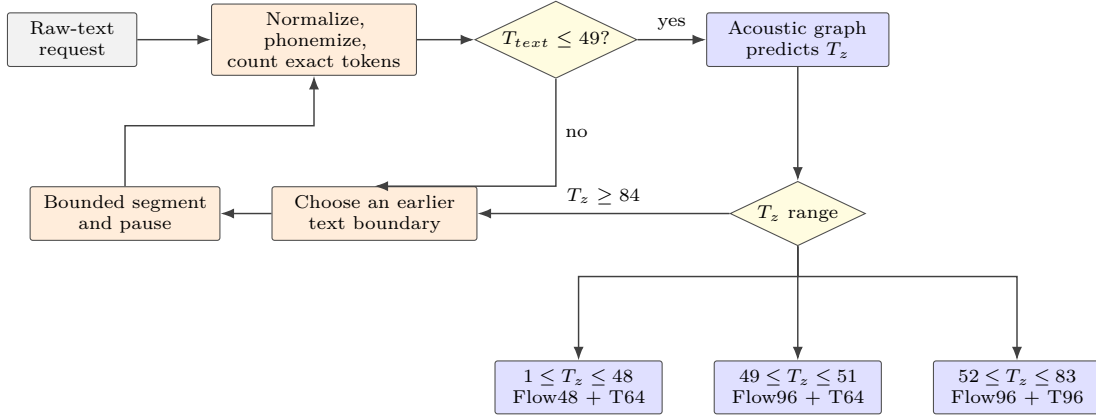


Figure 4: Shape-aware dispatcher and bounded long-text fallback. Token count is checked with the actual embedded frontend, while latent length is checked after duration prediction. Requests that exceed either static limit return to a valid text boundary before buffers are allocated.

Reverse flow had the same fixed-shape waste. Flow96 took about 465 ms for the 44-frame request. A Flow64 export reduced it to 308.3 ms. Flow48 reduced it to 234.6 ms and preserved exact FP32 valid-region equality against the masked 143-frame graph. Figure 4 shows the current dispatcher and the segmentation loop that precedes flow and decoder selection.

The route keeps flow and decoder limits independent. Flow96 occupies a separate 2 MiB flash partition; the main model package contains the acoustic model, Flow48, T64, and T96. Adding every historical model was rejected because it would exceed the 16 MiB flash device and retain unreachable choices.

4.6 On-device raw-text frontend

The release frontend normalizes English text and uses eSpeak-NG to produce North American English phonemes with stress and punctuation. The port embeds the exact eSpeak-NG 1.52.0 source [2]. A read-only 1 MiB FAT partition holds a stripped English voice, dictionary, phoneme tables, and intonation data. The full multi-language host data package was excluded.

The board frontend performs UTF-8 and punctuation normalization, release-specific word overrides, phonemization, lookup in the exact model symbol order, blank insertion, and bounds checking. The C++ normalizer implements the released order of money, date, time, phone, version, decimal, ordinal, number, acronym, punctuation, and model-specific rules without Python or regular expressions.

eSpeak exposed an internal-memory problem that model-only profiling would not find. A 16 KiB task stack overflowed; measured use exceeded 40 KiB. The final implementation creates a temporary worker with a 48 KiB internal-RAM stack, waits for completion, synchronously deletes the task, and frees its task control block and stack before neural inference. A dynamic-task version relied on deferred idle-task cleanup and reduced the largest free internal block to 31,744 bytes, preventing a second call. Explicit lifetime management restores the 63,488-byte largest block.

The stack cannot reside in PSRAM because eSpeak accesses flash-backed data and ESP-IDF forbids a PSRAM task stack on paths that can run while flash cache is disabled. Integration also found an unaligned embedded float asset. Replacing the generated embedded-file section with a four-byte-aligned assembly include placed the asset at a verified address ending in 518 and removed the fault.

The board passed nine release-parity cases. For every case, normalized text, IPA, each token ID, and token FNV-1a matched the pinned Python frontend. The corpus covers ordinary text, money, dates, time, identifiers, phone suffixes, versions, decimals, ordinals, long numbers, acronyms, street abbreviations, punctuation, and release-specific names. Board frontend time ranged from 2.455 to 6.099 ms with a mean of 3.839 ms. This cost is negligible relative to the learned path.

4.7 Bounded long-text planning

The static acoustic graph accepts at most 49 frontend tokens. Blank insertion means character count is a poor proxy, especially after number and date normalization. The firmware normalizes the complete input once and then asks the real board frontend to evaluate candidate boundaries. It prefers sentence boundaries, then sufficiently late clause punctuation, then whitespace. It never splits a UTF-8 sequence, a decimal point between digits, a word, a phoneme, or a token vector. An individual word that exceeds the bound returns an explicit error.

Each planned segment stores its source text, synthesis text, normalized text, phonemes, token IDs, boundary class, and following pause. The acoustic graph provides a second sizing gate: exact predicted durations are summed before the latent, flow, decoder, or PCM buffers are allocated. A typed duration overflow contains required and available frame counts. The planner can then move to an earlier word boundary. Runtime limits cap a request at 64 segments and 32 duration replans.

A narrow punctuation rescue avoids a one-word tail when only terminal punctuation exceeds 49 tokens. It may remove one final punctuation symbol from the synthesis text while preserving the original source boundary and pause in telemetry. Internal punctuation and punctuation on a fitting request remain unchanged.

The board verified planner decisions for short values, two sentences, normalized decimals, late comma selection, whitespace fallback, closing quote direction, and an oversized word. Successful segments contained at most 49 tokens; the longest planner-only request took 59.114 ms. A complete segmented request produced independently framed PCM segments, edge fades, and boundary silence. The host verified each segment before concatenating a listening WAV. The intended device implementation sends the same bounded segment directly to I2S/GDMA and reuses its inference buffers.

Long-text support bounds memory and preserves words. It does not carry prosody state across segments, and it cannot guarantee gap-free playback while the decoder is slower than real time. The suitable output is a sequence of short announcements rather than continuous narration.

5 Experimental Method

5.1 Measurement boundaries

Timers were placed around model loading, every learned graph, MCU glue, and the complete request. Binary PCM transmission begins after the inference timer. Reported model latency therefore excludes host-side WAV creation and serial transfer. Cold measurements include first model construction only when explicitly labeled. Warm measurements reuse resident objects.

Real-time factor is

$$\text{RTF} = \frac{t_{\text{compute}}}{N_{\text{samples}}/24000}, \quad (2)$$

where values above one are slower than playback. “Time to first audio” for the current graph equals the time to the complete PCM buffer because the decoder has no output callback or persistent streaming state.

The experiment did not collect large repeated timing distributions for every minor change. Each hypothesis used the cheapest board test that could change the implementation decision. Repeated runs were added when transport, scheduler variation, or a near-threshold result made a decision ambiguous.

5.2 Evidence hierarchy

Four forms of evidence answer different questions.

1. Float graph comparison verifies that a mathematical rewrite preserves the release model before quantization.
2. ESP-PPQ and embedded ESP-DL vectors verify quantized integer execution. The strongest parity tests compare every output element.
3. Framing, byte counts, board and host FNV-1a, PCM SHA-256, and WAV-header checks verify that captured audio is exactly what the board emitted.
4. Direct listening checks intelligibility, voice, stable noise, ending completion, and segmentation boundaries.

Table 3: Decoder optimization progression on the connected board. All rows use release weights unless the row explicitly changes static shape or precision.

Milestone	Decoder time	Audio geometry	Main mechanism
First complete decoder	12.809 s	T143, 1.525 s	Portable polyphase graph, stock scheduling
Stock graph, 512 KiB L2	10.078 s	T143, 1.525 s	Cache increase
Dual-core Conv1D and transpose	5.037 s	T143, 1.525 s	Time-axis split and specialized layout copy
Late-channel padding	3.971 s	T143, 1.483 s	24/12 channels padded to 32/16
Transpose fan-out CSE	3.833 s	T143, 1.483 s	Twelve exact duplicate copies removed
Slice-layout and resident service	2.986 s	T143, 1.483 s	NWC propagation and warm model objects
Final unpruned Full143 control	2.824–2.827 s	Segmented controls	Formula LUT, tiled transpose, mixed precision
T96 accepted route	2.013 s	Up to 83 valid frames	Short static decoder bucket
T64 accepted route	1.335 s	Up to 51 valid frames	Short-value decoder bucket

Passing an earlier level does not imply a later one. During development, a tiled decoder combined with reverse-flow pruning produced valid shapes, stable timings, correct checksums, and unintelligible noise. An earlier stack of runtime branch removals did the same. Both were rejected. The accepted control disabled all unvalidated pruning and tiling, reproduced an already heard short waveform byte for byte, and became the base for T64 and Flow48.

5.3 Audio transport

Early captures encoded PCM as hexadecimal UART records, doubling data size and making a 71 kB waveform take more than twelve seconds at 115200 baud. The persistent service uses framed binary PCM at 460800 baud. Each frame declares its byte count and ends with a board checksum. The host rejects a short or corrupt payload and retries transport only; frontend or inference failures are returned unchanged.

The CH340 serial bridge still lost bytes in a small number of large transfers. Splitting writes into 2 KiB pieces and waiting for UART completion improved capture reliability. This transport does not influence measured graph time. I2S/GDMA is the appropriate local audio sink because 24 kHz mono PCM16 requires only 48 kB/s and avoids diagnostic round trips.

6 Results

6.1 First complete waveform and full pipeline

The first complete decoder accepted the real [1, 128, 143] latent from the host golden and produced 36,608 samples. ESP-DL matched all 36,608 embedded INT8 outputs with maximum integer error zero. Decoder execution took 12.809 s for 1.525 s of audio, an RTF of 8.398. Model allocation was 245.59 KiB internal RAM and 5,153.09 KiB PSRAM. After model destruction, free PSRAM returned to its pre-load value.

The next milestone ran embedding, acoustic inference, duration/path/noise glue, inverse flow, decoder, and PCM conversion on the board. For **Testing one two three.**, the board produced 35,584 samples, or 1.482667 s. Board and host byte counts, chunk order, FNV-1a, PCM SHA-256, and the WAV header passed. The initial learned stages took 6.138 s, including 5.578 s in the decoder. A cold request took 7.411 s because models were loaded sequentially.

Table 3 summarizes the main runtime changes. Rows after the first three include graph, quantization, or shape changes and should be read as deployment milestones rather than a single-factor ablation.

6.2 Raw-text parity and synthesis

Table 4 gives four direct board frontend cases. The host sent only the text in the first column. The full nine-case normalization suite also matched every compared field.

The **Room 12.** synthesis request exercised normalization rather than the diagnostic frontend command. The board converted it to **Room twelve.**, produced 27 exact tokens, and returned 19,712 PCM samples, or 0.821333 s. Different raw inputs produced different token hashes, predicted durations, sample counts, and PCM hashes, excluding replay of a compiled waveform.

Table 4: On-device frontend parity examples.

Raw text	Tokens	Token hash	Time
Hello.	15	37b5b29c	2.361 ms
Good morning.	29	55879e9c	3.549 ms
Turn left.	25	132e7058	3.235 ms
Hello from the board.	43	380c159f	3.377 ms

Table 5: Final deployment resources after the relevant routes and frontend have been exercised. Decimal MB is used for allocator byte counts.

Resource	Measured value
Application image	1,933,440 B
Main four-member model package	8,409,632 B
Flow96 package	1,345,632 B
Fixed embedding/noise/token asset	86,198 B
Free bytes in main model partition	110,048 B
Free PSRAM	17,539,168 B
Largest free PSRAM block	17,301,504 B
Free internal RAM	99,551 B
Largest free internal block	63,488 B
Main service stack headroom	13,296 B

6.3 Final short-value route

The final connected-board gate used **Ready..** It produced 44 latent frames and 11,264 PCM samples, or 0.469333 s. Acoustic inference, Flow48, and T64 together took 1.656448 s. Flow48 took 234.609 ms and T64 took 1.335012 s. Flow48 reduced the learned path by 70.312 ms relative to Flow64 and by 12.104% relative to the earlier T64 plus Flow96 control. The current complete learned path has RTF 3.530 and throughput 0.283 times real time for this request.

Board and host agreed on PCM FNV-1a 51971333. The WAV SHA-256 is add6b5be...381b8; direct listening confirmed the spoken word, voice, and complete ending. This is a single acceptance gate, not a MOS study or a latency distribution.

The route is faster for short device values because its runtime follows the smallest valid static graph. A phrase such as **Battery low.** requires 80 frames and therefore uses Flow96 plus T96. Longer values may cross either the 49-token acoustic limit or the 83-frame deployment limit and are segmented at a text boundary.

6.4 Flash and memory

Table 5 records the final short-value build. The 16 MiB flash layout has a 4 MiB factory application partition, a 0x820000-byte main model partition, a 1 MiB G2P partition, and a 2 MiB Flow96 partition. About 852 kB remains after the last partition.

PSRAM is sufficient for the complete resident service. Internal RAM is the tighter product resource because it must also hold task stacks and any future DMA descriptors, network state, TLS buffers, and peripheral drivers. The 512 KiB L2 cache setting improves inference and removes on-chip memory from the heap. The final application retains a 64 KiB internal-memory reserve and measures the largest contiguous block rather than relying on total free bytes.

The current flash package leaves little room in its main partition. This is a deliberate final set of reachable routes rather than an archive of every tested bucket. A product requiring OTA slots, additional languages, or multiple voices would need a new partition table, external storage, smaller models, or a different decoder architecture.

6.5 Segmentation results

The connected planner kept a short screen value as one segment, preserved a sentence boundary between two statements, normalized a decimal before splitting, selected a useful late comma, split an unpunctuated sentence only at spaces, handled closing quotes, and rejected an over-budget single word. The longest accepted segment used exactly 49 tokens.

A complete multi-segment board request generated each segment, inserted its pause, returned framed PCM, and released bounded buffers for reuse. The final three-piece plan for one test message used 49, 41, and 49 tokens. The combined geometry was 96,544 samples, or 4.022667 s including pauses. An experimental decoder/flow combination produced only noise despite valid checksums and was rejected. Repeating the same plan with the unpruned exact-flow control preserved every planned source segment and the complete output geometry. Its short-path PCM was byte-identical to the previously heard baseline.

The automatic segmentation mechanism is therefore established at the text, token, duration, memory, and transport levels. The available listening evidence supports short independent values and the unpruned decoder control. The final long control still needs a formal listening study, and the model resets prosody at every segment. This distinction matters for publication: bounded long input is implemented, while continuous expressive narration is outside the present result.

7 Failure Analysis

7.1 Checksums cannot certify speech

Two optimization groups deleted decoder residual work at inference time, and one composite route combined a shortened decoder with a pruned inverse flow. The outputs were finite, non-silent, correctly sized, stable across runs, and transported without corruption. Listening found stationary noise with no intelligible words. The options were disabled in the accepted build.

This failure changed the experiment procedure. Approximate flow and decoder changes are now isolated, and each receives its own listening artifact before composition. A complete package cannot inherit quality acceptance from its members when they were calibrated separately. Output hashes establish reproducibility only within one declared graph and quantization state.

7.2 Quantization boundaries matter

Promoting a few sensitive residual convolutions to INT16 seemed attractive in a layerwise report. It worsened complete waveform error because the precision boundary fell inside a residual path. Promoting coherent front and output regions worked better. This supports block-level mixed-precision decisions for residual generators. Individual layer scores do not include the cost of conversion and requantization at branch merges.

The first all-INT8 output also demonstrated a presentation hazard. A file can be valid 16-bit PCM while its source tensor carries about eight effective bits. Container bit depth and model output precision must be reported separately.

7.3 Static shapes trade flash for response time

T64 and T96 reduce padded computation and preserve the released weights, but each duplicates most decoder parameters. The same applies to Flow48 and Flow96. Buckets consume flash and still return audio only after a complete graph. A bank of additional static lengths would eventually exhaust flash without producing native streaming. The final dispatcher uses two decoder lengths because both correspond to measured device-language cases and fit the available layout.

7.4 Memory lifetimes extend beyond tensors

The model fit in PSRAM early in the project. The difficult memory failures came from an eSpeak worker stack, deferred task cleanup, internal allocator fragmentation, cache allocation, model metadata, and an unaligned em-

bedded asset. Peak tensor estimates would not predict any of them. Reproducible MCU speech deployment needs measurements of internal free memory, largest blocks, worker stack high-water marks, resident model allocations, and state after repeated calls.

7.5 Serial output can distort perceived performance

Hexadecimal PCM at 115200 baud made a complete run appear much slower than inference. Binary framing and separate timers corrected the measurement, but the USB-UART bridge remained capable of dropping bytes. The host checksum turned silent corruption into an explicit transport failure. A local I2S sink removes this issue and uses little bandwidth compared with inference.

8 Streaming and Practical Use

The current decoder evaluates a complete static tensor and exposes PCM after its final operation. Bucketing lowers that wait for short utterances. It does not provide a sequence of valid early waveform blocks. An exact interior tile needs 13 latent frames of left and right context. If a valid tile contains C frames, it evaluates approximately $C + 26$ frames, so small tiles incur a large overlap tax. A 24-frame valid core evaluates 50 frames; a 64-frame core evaluates 90.

Throughput is the stricter constraint. The final `Ready.` route produces 0.469 s of audio in 1.656 s of learned computation. Starting playback from an early tile would eventually underrun. No buffering strategy can sustain playback when average production remains below one second of audio per second of wall time. A retrained causal or fixed-lookahead decoder, or an efficient spectral/recurrent vocoder, is required for conversational streaming.

The measured engine still has practical uses. A display can request speech for a selected value, menu item, timer, connection state, or sensor reading. The application can start synthesis when a value becomes likely to be spoken and play the completed short buffer on selection. A device can generate a bounded announcement before playback, cache common phrases, or queue several independent status messages. These cases benefit from local text handling, privacy, deterministic assets, and the absence of a network service. They can tolerate seconds of generation more easily than an interactive dialogue.

The system has one fixed male English voice. It does not provide voice cloning, speaker selection, multilingual inference, SSML coverage, native pitch control, or validated emergency/accessibility speech. eSpeak-NG is licensed under GPL-3.0-or-later; distribution of a proprietary firmware needs a licensing plan or a compatible replacement frontend. Inflect weights and code use Apache-2.0, ESP-IDF uses Apache-2.0, and ESP-DL uses MIT. License compatibility is part of deployment design, not a property of model weights alone.

9 Reproducibility

The public buildable source and evidence release is available at:

<https://github.com/inlanger/esp32-p4-inflect-tts>

The original paper-only snapshot is commit `336e7670a4e6a540d8841df35f8d9b3da3fae23f`. Its exact URL is:

<https://github.com/inlanger/esp32-p4-inflect-tts/commit/336e7670a4e6a540d8841df35f8d9b3da3fae23f>

The exact no-retraining firmware and artifact release is commit `f5701f9f92e92522086233d6c10879c07740b10f`. Supplemental sanitized decoder evidence was added in commit `d73897194ae034cf8a13d87add01ca15f99799f8`.

The deployable ESP-DL model package is distributed separately at:

<https://huggingface.co/inlanger/inflect-nano-v2-esp32p4-espdl>

Its immutable Hub revision is `282a7eaf953e8a9e972b86d844d1d2823a48df21`. The Hub repository contains only model packages, fixed assets, a retained WAV, the model card, manifest, and model-license notices. Firmware, the ESP-DL patch, raw JSON telemetry, build scripts, and supplemental analysis remain in the GitHub repository. Experimental retrained G5–G18 decoder lineages are excluded from both releases.

The final main package is named `short64-flow48`; it contains the acoustic graph, T64 and T96 waveform decoders, and the CSE Flow48 graph. Its size is 8,409,632 bytes, file SHA-256 is `3a267b2310a36ad5cc561e5ac3da78ee89aec66fb3b4c7df3abbd43280304bb9`, and its internal package digest is `b3ffb16b34e180a040796727385500adc6ad721d64c3b0082fdad25479c843cb`. The matching application SHA-256 is `50462fa0add9f59b68ca9b8e7adc8525e4eb4427d839405694652a101b9d1634`.

The final short board artifact is stored with machine-readable telemetry. Its WAV SHA-256 is `add6b5be3e21278a46837c5a0269cee9392bbf6dc6081327082797963fd381b8`; the corresponding JSON SHA-256 is `17e17cb633b26863ae7c990cae0e0529f4fcec9c5917ffdca7eaa8b21500ec31`. The public supplemental record at `evidence/decoder-surface/decoder-surface.json` contains every decoder convolution shape, nominal MAC count, real activation statistics, parameter group, and receptive-field result. Only two workstation-local paths were removed from the retained research record; all numeric evidence and source hashes are unchanged. Its public SHA-256 is `ad3fe556db6ea0ca54115ad7413ff1d7a88d7df18ef0f1c81e26419180c254bb`.

Reproduction requires the early P4 revision or a deliberate retarget. The checked-in defaults select revision 1.x and 360 MHz; a binary built with these settings is not a revision-3.x image. Board timing also depends on 200 MHz hexadecimal PSRAM, cache size, parameter-copy policy, compiler optimization, and the local ESP-DL Conv1D, transpose, and activation changes.

10 Limitations and Publication Scope

The paper reports one physical board and one fixed English voice. There is no formal MOS, MUSHRA, or multi-listener preference study for the embedded outputs. Listening decisions were direct engineering gates. The upstream model’s published quality metrics describe its host FP32 release and cannot be assigned to this mixed-precision P4 port.

Power draw, regulator and SoC temperature, long-duration throttling, network coexistence, I2S playback, watchdog behavior under a full application, and 24-hour stability have not been measured. Internal RAM is low enough that a network stack and audio DMA require a fresh budget. The current USB protocol is a diagnostic adapter rather than a product interface.

The static acoustic graph accepts 49 tokens. Segmentation extends finite input length by issuing independent bounded requests. It does not model prosody across boundaries. Natural duration-overflow recovery is implemented and bounded; the test corpus did not produce a normal phrase that triggered it after the token gate, so that particular replan branch lacks connected-board coverage.

Claims of priority should remain narrow. We did not find an official or open Inflect-Nano-v2 deployment on ESP32-P4 during the literature and repository search. Absence from a search is not proof that no private or unpublished port exists. The publishable contribution is the documented complete system, measured runtime work, and evidence procedure.

An arXiv systems report is justified by the complete deployment and the operator-level findings. A stronger peer-reviewed submission would add a several-board replication, controlled listener ratings, power measurements, I2S playback, and comparisons against at least one purpose-built MCU vocoder. Achieving faster-than-real-time generation with a trained causal or spectral decoder would constitute a separate model result.

11 Conclusion

We deployed a complete raw-text neural speech synthesizer on an ESP32-P4. Normalization, English phonemization, tokenization, acoustic inference, duration and latent construction, inverse flow, waveform decoding, and PCM16 conversion execute on the microcontroller. The host supplies text and receives diagnostics; it performs no speech computation.

The result depended on system work beyond model compression. Exact polyphase upsampling removed wasteful zero insertion. Dual-core Conv1D scheduling, channel padding, layout specialization, graph CSE, mixed precision, resident models, and shape-aware routing reduced the first 12.809-second decoder to a 1.335-second T64 path for a 44-frame short value. The complete learned route takes 1.656 seconds and fits with the frontend in the board’s flash and PSRAM. The accepted build retains recognizable, intelligible short speech and rejects faster candidates that produced noise.

The current engine is a complete local TTS implementation for short device speech and bounded announcements. Its decoder is slower than real time and returns complete buffers. These limits are measured properties of the released model and runtime, and they identify the next technical problem: a decoder trained for MCU throughput and explicit streaming state.

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